PySpark Tutorial for Beginners

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Agenda for Today's Session

01 Get a brief understanding of what exactly is PySpark

Introduction to PySpark

02 Understand how PySpark is used in the industry

PySpark in Industry

03 Learn how to install PySpark on your machine

PySpark Installation

04 Learn about various PySpark fundamentals like sparkcopntext, sparkconf, RDDs, SparkFiles, etc.

PySpark Fundamentals

05 Perfom a hands on for PySpark

PySpark Demo

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Introduction to PySpark

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What is PySpark?

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What is PySpark?

Python is a powerful high-level, object-oriented programming language with simple easy to use syntax.

Apache **Spark** is open source, general-purpose distributed computing engine **used** for processing big data.

PySpark

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What is PySpark?

PySpark

"PySpark is the collaboration of Apache Spark and Python. It is a python API for spark which easily integrate and work with RDD using a library called Py4J."

Note: It allows the developers to write driver programs in python

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Advantages of PySpark

Speed

100x faster than traditional large-scale data processing frameworks.

Powerful Caching

Simple programming layer provides powerful caching and disk persistence capabilities.

Real Time Computation

There are many variations of passages but the majority have alteration in some form by injected are many variations of .

Deployment

Can be deployed through Mesos, Hadoop via Yarn, or Spark's own cluster manager.

Polyglot

Supports programming in Scala, Java, Python, and R.

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When to use Python or Scala with Spark?

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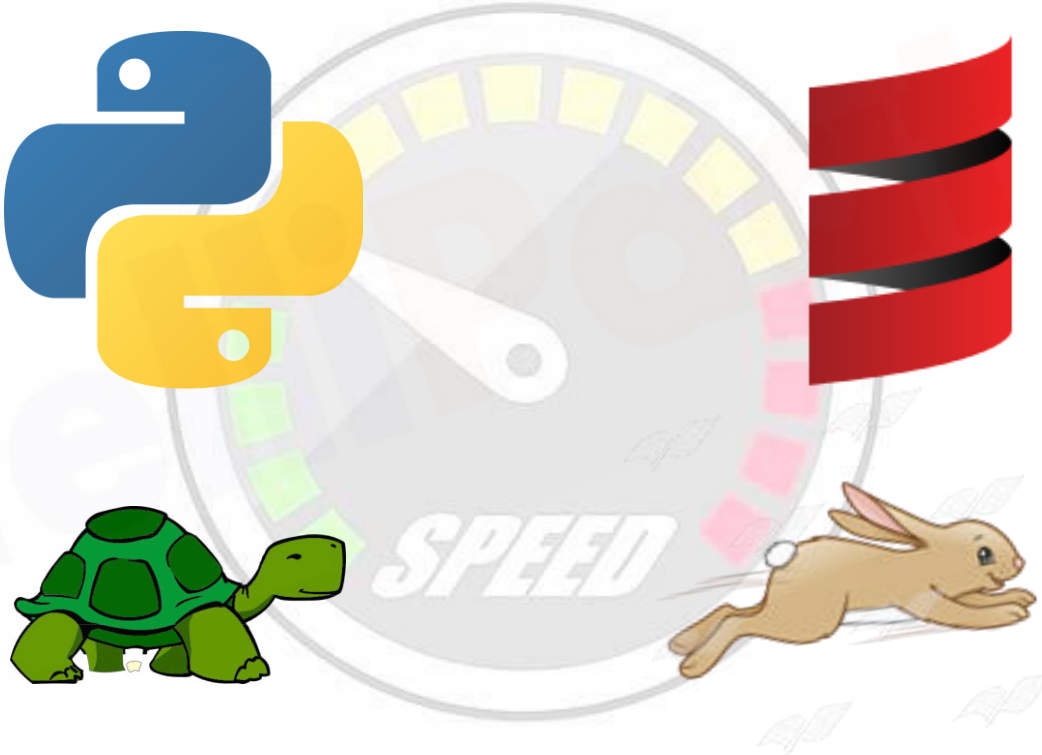
Python vs Scala

- Performance speed
- Learning Curve
- Data science Libraries
- Complexity

Performance Speed



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Learning Curve

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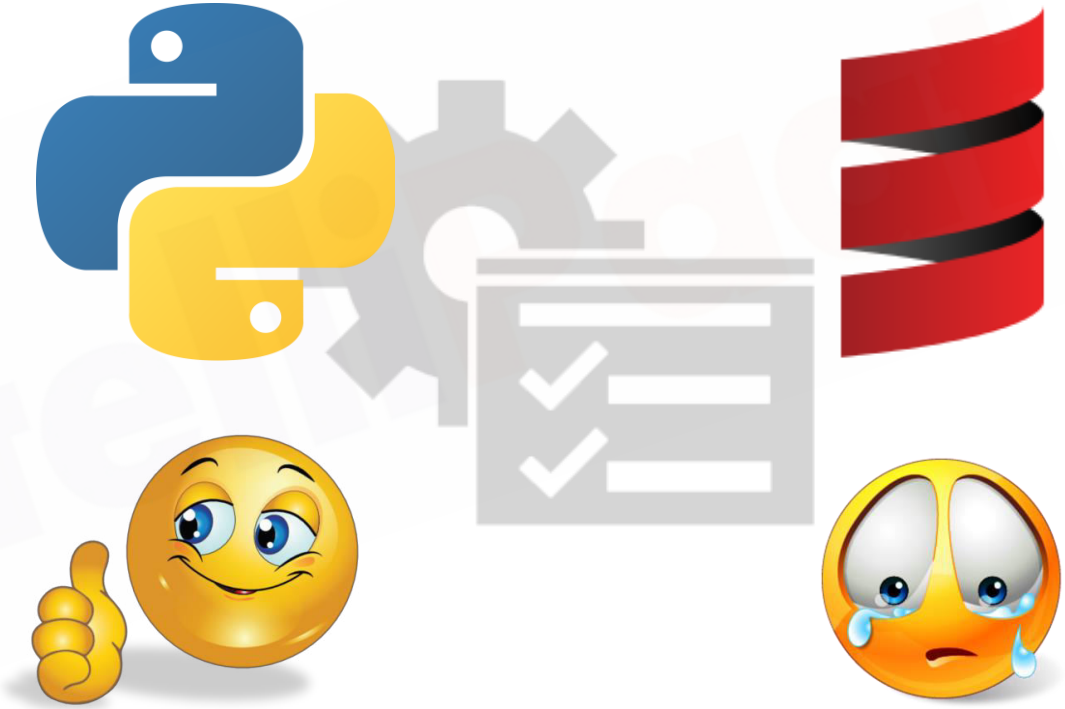
Data Science Libraries



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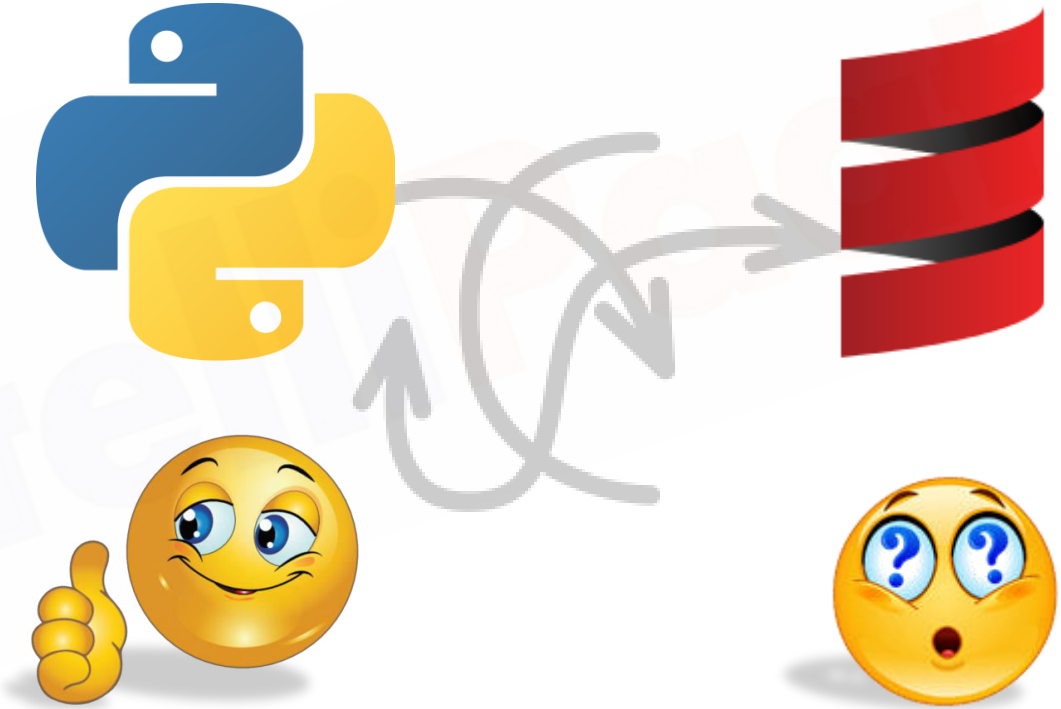
Complexity



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PySpark in Industry

YAHOO!

NASA

Alibaba.com

tripadvisor

ebay

NOKIA

NETFLIX

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PySpark Installation

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PySpark Installation

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Pyspark Fundamentals

MLlib
Machine Learning API of Spark which in turn supports various ML Algorithm

04

Dataframe
Distributed collection of rows under named column

RDDs
Stands for Resilient Distributed Dataset and is the building block of every spark application

03

02

SparkFile
Helps in resolving the path of files added to spark

SparkConf
Provides configuration to run a Spark application on a local machine or cluster

01

SparkContext
SparkContext is the entry gate for any spark derived application or functionality.

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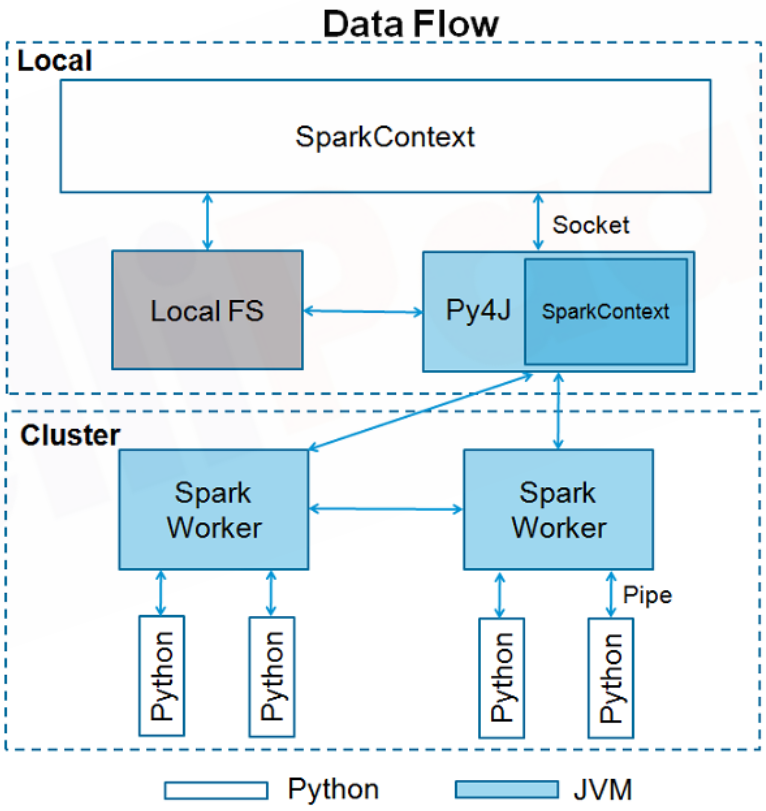
SparkContext

Entry point to any spark functionality

What is SparkContext?



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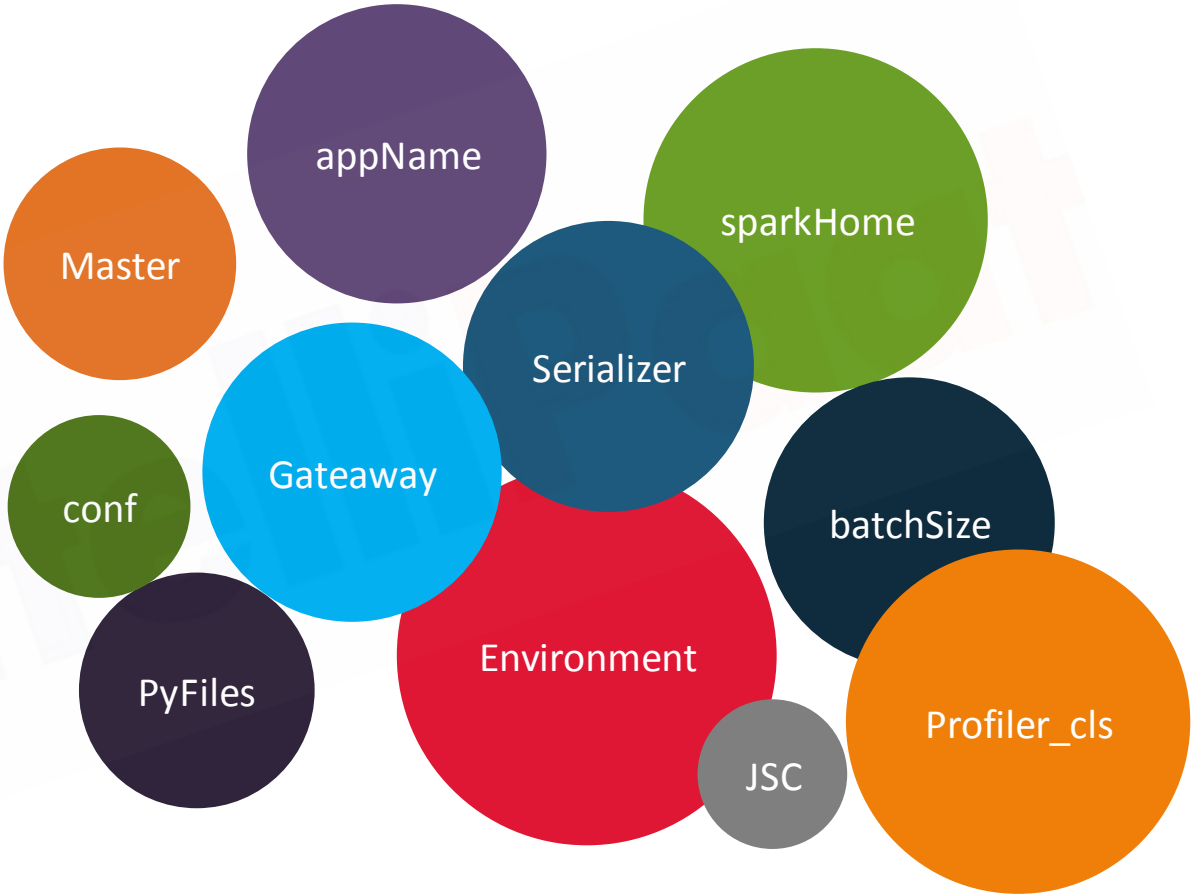
SparkContext : Parameters

Entry point to any spark functionality



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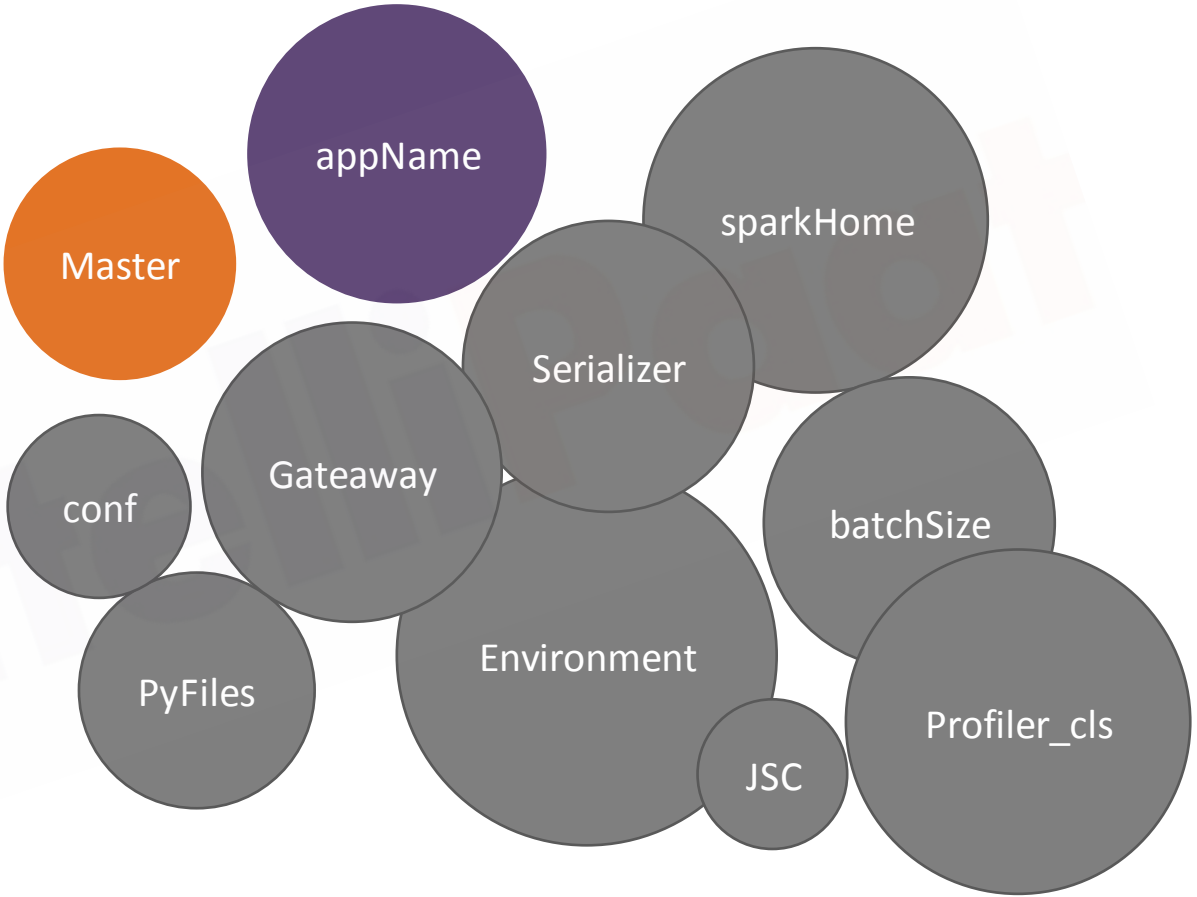
Entry point to any spark functionality



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SparkConf: About

Provides configuration to run a Spark Application on a local system or a cluster

What is SparkConf?

- Provides configurations to run a Spark application
- Sparkconf object is used to set different parameters which takes priority over the system properties
- Once the object is passed to spark it becomes immutable

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SparkConf: Attributes

Provides configuration to run a Spark Application on a local system or a cluster

Most commonly used attributes

- **set(key, value):** To set a configuration property
- **setMaster(value):** To set the master URL
- **setAppName(value):** To Set an application name
- **get(key, defaultValue=None) :** To Get a configuration value of a key
- **setSparkHome(value):** To Set Spark installation path

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SparkConf: Output

Select C:\WINDOWS\system32\cmd.exe

```
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 2.3.2
      /_/

Using Python version 3.7.1 (v3.7.1:260ec2c36a, Oct 20 2018 14:05:16)
SparkSession available as 'spark'.
>>> from pyspark.conf import SparkConf
>>> from pyspark.context import SparkContext
>>> conf = SparkConf().setAppName("PySpark Demo App").setMaster("local[2]")
>>> conf.get("spark.master")
'local[2]'
>>> conf.get("spark.app.name")
'PySpark Demo App'
>>>
```

SparkConf: Example

Provides configuration to run a Spark Application on a local system or a cluster

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SparkFile

Resolves the path of files added through `SparkContext.addFile()`.

- **`get(filename)`:** specifies the path of file added through sc.addFile()
- **`getrootdirectory()`:** specifies the path of file root directory added through sc.addFile()

C:\WINDOWS\system32\cmd.exe

```
        at scala.Option.map(Option.scala:146)
        at org.apache.spark.deploy.SparkSubmit$.doPrepareSubmitEnvironment(SparkSubmit.scala:415)
        at org.apache.spark.deploy.SparkSubmit$.prepareSubmitEnvironment(SparkSubmit.scala:250)
        at org.apache.spark.deploy.SparkSubmit$.submit(SparkSubmit.scala:171)
        at org.apache.spark.deploy.SparkSubmit$.main(SparkSubmit.scala:137)
        at org.apache.spark.deploy.SparkSubmit.main(SparkSubmit.scala)
2018-11-10 18:39:23 WARN  NativeCodeLoader:62 - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 2.3.2
      /_/

Using Python version 3.7.1 (v3.7.1:260ec2c36a, Oct 20 2018 14:05:16)
SparkSession available as 'spark'.
>>> from pyspark import SparkFiles
>>> from pyspark import SparkContext
>>> path = os.path.join("/Users/intellipaat/Downloads/spark-2.3.2-bin-hadoop2.7", "Fortune5002017.csv")
>>> sc.addFile(path)
>>> SparkFiles.getRootDirectory()
'C:\\Users\\intellipaat\\AppData\\Local\\Temp\\spark-3d68e97a-8772-4f50-bcaf-ef5036732b1f\\userFiles-a787d391-9a04-49fa-bda5-65745eb22ffa'
>>> 
```

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Resilient Distributed Dataset

Fundamental data structure of Apache Spark

What is RDD?

Resilient – if the data in the memory is lost, it can be recovered (fault tolerant)

Distributed – Data is chunked into partitions and is stored in memory across the cluster

Dataset – Initial data can come from a file or can be created using the program

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Operation in RDDs

Types of Operation in RDD

- Transformation
- Action

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Operation in RDDs

- Transformation
- Action

What is Transformation in RDD?

Used to create new RDD either by applying the operation on an existing RDD or by making entirely new RDD

- map
- filter
- distinct
- reduceByKey
- mapPartitions
- sortBy

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Operation in RDDs

- Transformation
- Action

What is Action in RDD?

Applied directly on the datasets to perform certain computations on the datasets

- take
- first
- countByKey/countByValue
- collect
- collectAsMap

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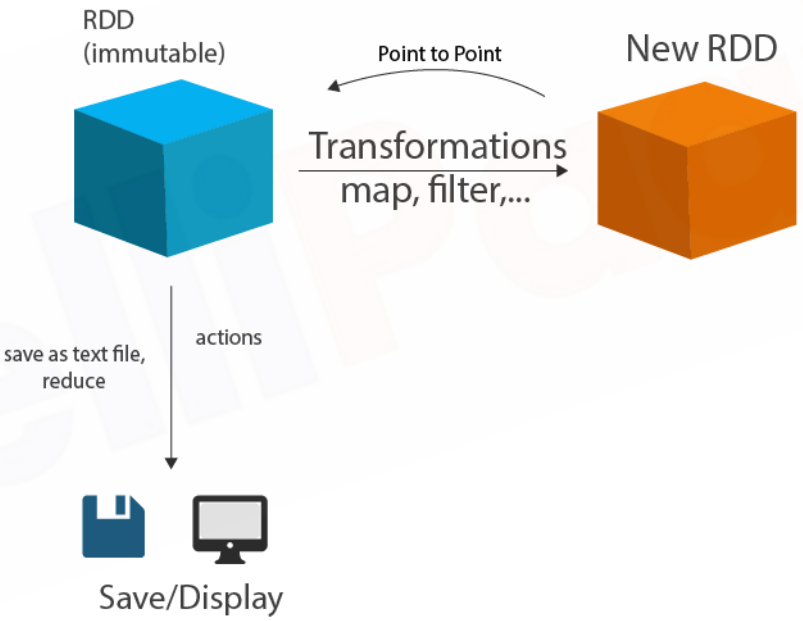
Action & Transformation



Operation in RDDs

- Transformation
- Action

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RDD: When to use?

When to use RDD?

Low Level transformation

you want low-level transformation and actions and control on your dataset;

Data manupulation

you want to manipulate your data with functional programming constructs than domain specific expressions;

No Schema

you don't care about imposing a schema, such as columnar format, while processing or accessing data attributes by name or column

Unstructured Data

your data is unstructured, such as media streams or streams of text;

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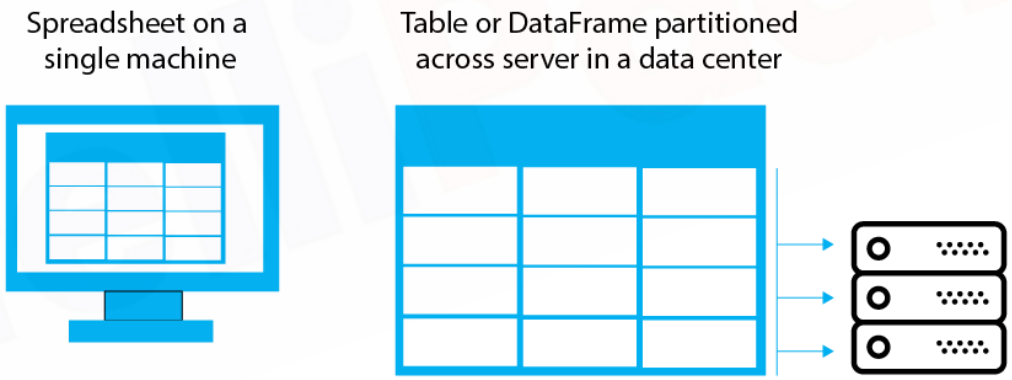
Dataframe

What is a DataFrame?

A structured API which represents a table of data with rows and cloumns



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RDDs vs Dataframe

Difference between RDD & Dataframe

Parameter	RDD	Dataframe
Release	Since Spark 1.0	Since Spark 1.3
Data Representation	Elements spread across the cluster	Organized into named columns
Optimization	No inbuilt optimization engine	Optimization takes place using catalyst optimizer.
Schema Projection	Defines the schema manually	Auto discovery of the schema

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MLlib

What is MLlib?

A machine learning library that supports various machine learning algorithms.

- Classification
- Regression
- Clustering
- Collaborative Filtering
- Dimensionality Reduction

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